Counterplay CP-01: A failed feasibility screen for matched critical-reply repairs

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Abstract

An intervention study needs an eligible population before it needs a trained model. We examined the feasibility of CP-01, a proposed audit comparing equal, same-direction scalar corrections at critical and noncritical opponent replies inside exact-rule adversarial search. A first development screen and one prospectively reviewed amendment reused the same 164 behavior episodes in Connect Four 6x7 and Reversi6. Every attempted terminal-solved reference fork completed: 57 in the first screen and 59 in the amendment. Neither screen yielded a repairable critical-reply root in Connect Four; Reversi yielded 11 of 50 roots, only four with potential nonzero signed-error common support. The revised design therefore failed its frozen eligibility gates and stopped before trainer construction. We explain a support constraint: with predictions bounded in [-1,1], reference values -1 and +1 admit no shared strictly positive or strictly negative residual stratum. We report finite-screen counts, dependence, costs and reconstruction evidence. These are results about this audit design and sampled population, not about JEPA playing strength or the absence of critical replies in either game.

1. Motivation and scope

Predictive models used in planning must preserve distinctions that affect decisions. In an adversarial fork, one opponent reply can determine an own action's minimum utility. An average prediction error need not emphasize that reply. CP-01 proposed to examine whether the location of an error matters after matching its magnitude and correction direction within a checkpoint. The intended empirical contribution was a common-utility, complete-fork audit, not a new representation principle, architecture, or superiority claim.

This report asks a narrower question: does the declared development population support that proposed matched intervention under its actual rules? It records the prescribed feasibility stop. The learned-model study was not executed. No loss, regret, reply-preservation, match score, Elo, or model-effect estimate is available. No model-comparison null hypothesis was tested.

2. Related work and claim boundary

MuZero establishes learned-model game planning; Value Equivalence establishes the importance of preserving quantities used by value planning. CP-01 does not claim either idea as new.

AD-WM, section III-E, measures candidate ranking and elite-selection regret in each model's planning space. Counterfactual Action Evaluation, section 4.2, traces exact-state action interventions through observations, embeddings and predictions. Dual-Frontier, appendix A.1, specifies which operator, action, budget and randomness must remain fixed under oracle replacement. One-Step Prediction Fidelity, section 3.3, uses paired planning and matched continued-training controls when examining prediction/planning dissociation.

These precedents limit any claim that counterfactual forks, decision-relevant error, oracle substitution, or a negative prediction/planning association is itself novel. CP-01's candidate distinction was the operational definition of opponent-reply criticality and matched repairs on common terminal utilities. The present evidence does not validate that distinction as an effective diagnostic. The source comparison is targeted, not an exhaustive novelty review; no methods-paper

priority or publication-readiness claim is made.

3. Prospective design and feasibility methods

3.1 Forks, criticality and the intended intervention

For root state s, own action a, and opponent reply b, let q(s,a,b) be terminal minimax utility for the root player after those two exact legal transitions. Utilities are -1, 0, or +1. Define Q(s,a) = min_b q(s,a,b), and V(s) = max_a Q(s,a). A terminal own action has exact utility without a reply set. Regret of a selected action is V(s) - Q(s,a_selected), in [0,2]. Only terminal-complete references are used; no fixed-horizon value is called full-game regret.

A critical reply is a unique worst reply to a suboptimal action, separated from the next-worst reply by at least 0.5, whose deletion would raise that action's minimum to at least the original root optimum. The primary candidate also requires a nonterminal critical child and a nonterminal, noncritical control under the same own action. Lexicographic selection occurs using references before model scores. Terminal replies remain exact and are not repairable learned entries. A forced Reversi pass consumes a ply and switches the actor; terminal detection precedes expansion.

The proposed primary intervention would replace one predicted scalar q_hat with its common reference value. Its within-checkpoint contrast was regret after noncritical repair minus regret after critical repair, restricted to matched checkpoint/root units in multi-step JEPA. Controls would match residual sign and absolute-error magnitude within 0.1. Ties use action IDs; the zero/near-zero residual stratum has a 1e-10 tolerance. Match coverage is part of the estimand, not a removable inconvenience. A zero-match or unresolved seed would make the equal-seed primary estimate missing. No such model estimate was computed here.

Five arms were contemplated: multi-step and H1-only JEPA, value-only latent rollout, decoded-state rollout, and exact-state direct-leaf. The value-only latent arm was the relevant identical-path auxiliary-objective control; direct and decoded paths were intentionally different and could not by themselves identify a JEPA effect. Behavior-outcome supervision was distinct from minimax references. Re-encoding and reference substitution would be privileged descriptive diagnostics, not proof of information loss. No final seed/count panel was frozen, and none of these fits ran.

3.2 Episodes, windows, isolation and references

The development recipe generated 32 training and 50 development episodes per game. Episode-index cycling assigned four ordered pairs of uniform and tactical policies. Tactical play selected an immediate win if available, otherwise a move without an immediate losing reply when possible, otherwise a legal move; ties were uniform. The split/game/episode seed came from a fixed SHA-256 namespace. Every generated state, legal set and successor was compared with the separately implemented bitboard rules. This is same-project independent code, not a third-party engine validation or formal proof.

Each selected training window contained four legal transitions ending nonterminal, supplying valid H1/H2/H4 targets for every arm. Short/terminal windows were counted and excluded. Canonical path/content duplicates were collapsed with source multiplicities retained. Selection was by hash, not outcome. The bank contained 128 unique windows/game for training and 32/game for development. Evaluation root canonical keys were excluded from all selected root and future-state targets. This is selected-position isolation; common openings and unmeasured reference-descendant overlap preclude a whole-tree or complete state-disjointness claim.

Each development episode contributed at most one nonterminal late root with at least two own actions, selected by minimum hash before solving. The first screen allowed at most eight empty cells in Connect Four and six in Reversi. No candidate, duplicate, failed reference, or ineligible

root was replaced. The independent negamax solver completed every legal two-ply fork to terminal minimax values or rejected the entire root. The cap was 100,000 solver nodes and two wall seconds per root. The node convention counts solver visits, not every outer exact-rule call; wall/CPU measurements include construction.

3.3 Gates, amendment and costs

The initial engineering gate required at least five reference-pairable roots per game and at least 90% reference completion. These are minimum feasibility floors, not power or representativeness guarantees. After the first failure, an independent scientific review approved one final prospective revision: Connect Four's empty-cell limit increased from eight to twelve, with the original episodes, seed namespace, pre-solve selection and solver caps fixed. Reversi remained unchanged. Source and amended protocol were pushed and verified before invocation.

The amendment additionally required at least two roots/game and at least 20% of pairable roots with potential nonzero signed-residual common support. This tests only domain possibility, not actual model matching. The first batch's CPU time was subtracted from a cumulative 300 builder-CPU-second cap; an external 330-second wall deadline bounded each invocation. A 2 GiB address space limit applied to the data producer. Failed gates remained durable events. The amendment permitted one invocation, with an explicit no-further- screen stop implemented by the operator/protocol rather than a software lock.

4. Findings

4.1 Reference and support yield

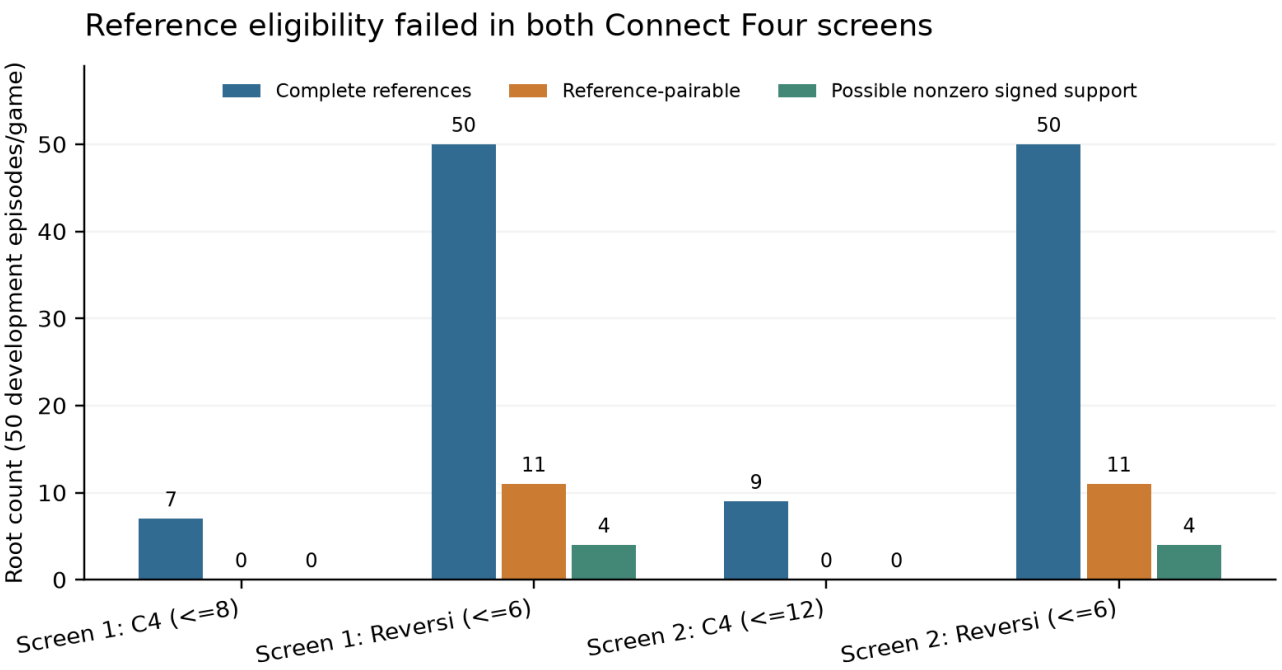
Table 1. Finite development-screen counts; screens reuse the same episodes.

Screen	Game	Episodes	No candidate	Complete / attempted	Pairable	Potential nonzero support
1	connect4-6x7	50	43	7 / 7	0	0
1	reversi6	50	0	50 / 50	11	4
2	connect4-6x7	50	41	9 / 9	0	0
2	reversi6	50	0	50 / 50	11	4

Potential support was diagnosed after screen 1 and frozen as a necessary gate before screen 2. It is not actual model matching.

Table 2. Builder resources; excludes tests, agent work and rendering.

Screen	Wall seconds	CPU seconds	Peak RSS bytes	Windows train / development
1	15.551	15.392	80,281,600	256 / 64
2	16.308	16.143	80,220,160	256 / 64



Reference-pairability and possible nonzero signed support in the two dependent screens.

All attempted references completed, so the failed gate cannot be attributed to solver timeouts in this batch. In the first Connect Four screen, five of seven roots had all own-action reference minima tied. Its sole root-decision- critical action had a terminal critical child, which the study correctly treated as exact rather than repairable. Widening the window admitted nine roots but still no eligible matched-repair population. This does not show that earlier or differently sampled Connect Four positions lack nonterminal refutations.

Both screens retained 256 training and 64 development windows and passed that count gate. The first bank's prohibited selected-state/root collision count was independently recomputed as zero. Exact decoded episode-list comparison verified that all 164 episodes were identical across screens. There were not 328 independent episodes or two independently generated replication corpora. Window integrity and reference completion did not rescue scientific eligibility.

4.2 A bounded-value constraint on signed matching

Let predictions lie in [-1,1] and let e = q_hat - q_ref. When the critical reference is -1, e_critical is nonnegative. When its control reference is +1, e_control is nonpositive. They cannot both be strictly positive or both be strictly negative, regardless of training. At exact zero, a scalar correction does nothing. With the declared numerical tolerance, both can enter the zero/near-zero stratum; that is not evidence of nonzero positive/negative common support, and a tiny correction near a decision tie need not be harmless. We make no claim that this stratum was observed or has a measured effect.

The first Reversi screen contained twelve critical=-1/control=+1 branch pairs under its selected reference candidates, and six critical=0/control=+1 pairs. Only four roots had any control with potential nonzero same-sign domain support. A domain-possible pair may still fail the 0.1 caliper or actual model coverage. This algebraic constraint explains why a large reply bank or additional training is not sufficient to guarantee this primary estimand. It is a property of the proposed diagnostic, not a new theorem about JEPA.

4.3 Stop disposition and unavailable outcomes

The final screen failed both reference-yield and necessary-support gates in Connect Four. The frozen stop rule retired the two-game mechanism design before trainer construction. No third root screen ran. No development model pilot, final panel, locked evaluation, second training corpus, paired match, or chess extension was executed. The earlier 100-fit planning envelope was not an executed or finally scheduled experiment. Loss/regret/Elo results remain unavailable, not zero. Review 2 did not approve pilot readiness, and the locked-study Review 3 was not reached.

The two builders consumed 31.534 CPU seconds in total, excluding mathematical tests, agent reasoning, packaging and report rendering. No rental was initiated. Actual agent tokens and billing were not exposed and remain missing. A monthly $50 compute ceiling is a budget, not measured expenditure. These distinctions prevent a cheap feasibility screen from being presented as a complete study's compute or agent cost.

5. Discussion and limitations

The useful conclusion is that CP-01 v3 should not be trained as specified: its chosen Connect Four population supplies no eligible primary units, and even Reversi's reference-pairable population substantially overstates possible strict-sign support. This validates the decision to screen the diagnostic before implementing a trainer; it does not validate the diagnostic's scientific mechanism or favor any model family.

Eligibility depends jointly on root sampling, uniqueness of the worst reply, root-level deletion criticality, exact terminal overrides, same-action nonterminal controls, and residual matching. Rare eligibility is therefore not evidence that opponents rarely have decisive replies. Removing an eligibility restriction after seeing these screens would change the question. A Reversi-only design, controlled perturbation study, or broader root generator could be investigated prospectively, but none is silently substituted here.

The recipe is small, same-size, policy-dependent and late-game. There are no population prevalence intervals, power estimates or general board-game claims. The two screens are dependent and the descriptive report scope was adopted after development observations; it is not a confirmatory preregistration. The mathematical support check is necessary, not sufficient. Project-owned independent rules can share errors; tractable differential fixtures do not prove every reference. Timing is local to the documented host, and peak RSS is the observed process statistic, not a complete memory history.

Foundation work separately repaired a covariance-gradient weighting mismatch: all six affected arms and every parameter tensor passed 192 synthetic numerical checks after 35 baseline failures. That engineering result does not constitute learning or model-performance evidence. Integrated training, optimizer/EMA/RNG resume, production measurement cadence, scalable exports, inferential analysis calibration and independent final evaluation remain prerequisites for any future learned-model study, rather than completed features of this report.

6. Reproducibility and evidence boundaries

The repository preserves both prospective protocol versions, the reviewed amendment, source identities, every failed gate, compact per-root rows, generated tables/figure and the claim map. Detailed commands are in REPRODUCE.md. The report's tables and plot can be regenerated from published receipts and unit CSVs without raw boards or fitted weights. The two data-building source commits and exact data SHA-256s identify optional raw regeneration in separate checkouts; reruns are recovery attempts, not permission to extend this stopped study.

GitHub is the selected off-device destination. Raw episodes, windows, checkpoints, environments and logs remain local and ignored. A source hash does not reconstruct missing bytes by itself. A fresh clone of the documented earlier milestone passed ten focused tests using the same host and existing environment; this was not replacement-machine recovery or recovery of raw data. The final package's scoped review/recovery receipt identifies what was actually verified. The PDF is an automated project technical report, not a submitted paper or a claim of acceptance-ready novelty.

7. Conclusion

Two finite, dependent screens failed the declared population gates for the CP-01 two-game matched critical-reply study. Complete references and usable training windows did not imply that the primary matched population existed. The audit stopped before model training, preserved its unsuccessful attempts, and exposed a bounded-value/sign-support constraint that must be considered in any redesign. No conclusion about JEPA effectiveness follows.

References

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